Week 35 Draft: Affinity-Model Architecture and RFdiffusion–ProteinMPNN Binder Design

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Executive summary

This report will address two connected but scientifically distinct workflows. The first is **affinity prediction**: Boltz-2, Nesso-1, and FlashBind map protein–small-molecule information to an affinity score, with different use of sequence, MSA, pocket, and three-dimensional coordinates. The second is **protein-binder design**: RFdiffusion proposes a binder backbone around a target protein and ProteinMPNN assigns amino-acid sequences compatible with that backbone [1, 2].

RFdiffusion and ProteinMPNN are included because they expose a useful location for adding physical information during design. They are not affinity models, and their native scores are not binding free energies or probabilities of experimental success. I will keep their analysis separate from the small-molecule affinity benchmark and use the audited insulin-receptor example as a concrete worked case.

1 Questions for the week

1. What tensors and geometric representations connect the input of each affinity model to its affinity head?
2. What exactly do RFdiffusion and ProteinMPNN receive and return?
3. How do hotspot conditioning and stochastic sampling alter the proposed binder backbone?
4. Where could a physics-aware interface score enter the binder-design workflow without being confused with ProteinMPNN likelihood?

2 Two different prediction problems

Task	Representative input	Native output	Appropriate evaluation
Small-molecule affinity prediction	Protein sequence or structure and ligand chemistry/pose	Affinity score, and for Boltz-2 a cofolded complex	Within-target affinity correlation; protein and ligand structure recovery when references exist
RFdiffusion binder generation	Target coordinates, contig, optional hotspot residues, checkpoint, and random seed	Target plus a generated binder backbone and reverse-diffusion trajectories	Backbone geometry, target-relative pose, contacts, clashes, diversity, and downstream validation
ProteinMPNN sequence design	Fixed target/binder backbone, design mask, checkpoint, temperature, and seed	Amino-acid sequences, per-residue probabilities, and sequence-model scores	Sequence diversity, structural compatibility, independent complex prediction, physical interface checks, and experiment

Table 1: The native outputs answer different questions. In particular, ProteinMPNN likelihood is not an affinity, and RFdiffusion confidence is not a binding measurement.

3 RFdiffusion: target-conditioned backbone generation

3.1 Input and output

In the binder-design setting, RFdiffusion receives a three-dimensional target structure and a contig specification that identifies fixed target residues and the length or length range of the new chain. Optional hotspot residues nominate the target region where an interface should be formed. The checkpoint and random seed are also scientifically important inputs because different checkpoints encode different training tasks and different seeds generate different samples.

Only the binder is diffused in the target-conditioned protocol. The target coordinates remain fixed and provide conditioning throughout reverse diffusion. A simplified input–output statement is

$$(\text{target coordinates},\ \text{contig},\ \text{hotspots},\ \text{seed}) \longrightarrow (\text{target} + \text{binder backbone},\ \text{trajectory}).$$

The generated chain is initially represented as a backbone proposal; it is not a chemically complete protein and it has no designed amino-acid sequence.

3.2 Why equivariance matters

RFdiffusion represents protein geometry using residue-level positions and orientations. If every input coordinate is transformed by the same rigid motion,

$$\mathbf{x}_i' = Q\mathbf{x}_i + \mathbf{a},$$

the denoised output should transform by the same Q and $\mathbf{a}$. This SE(3)-equivariance prevents the model from attaching meaning to the arbitrary orientation of a protein in a coordinate file. It also explains why geometric guidance should depend on invariant distances, angles, contacts, or energies, rather than laboratory-frame Cartesian coordinates.

3.3 What hotspot conditioning establishes

Hotspots are conditioning information, not residues that RFdiffusion discovers as experimental binding sites. In the existing matched insulin-receptor ablation, both arms used the same complex checkpoint, target, contig, ten seeds, and sampling settings; only the hotspot tensor was removed. The guided backbone had a smaller minimum Cα distance to A59, A83, or A91 in 9 of 10 matched seeds. The median minimum distance was 6.38 Å with hotspots and 9.72 Å without them.

This demonstrates that hotspot information steers backbone placement. It does not demonstrate affinity, folding, specificity, or validated pocket discovery. The ten guided seeds also produced diverse backbones: pairwise intrinsic trace RMSD had a median of 13.36 Å, and target-aligned binder-pose Chamfer distance had a median of 5.08 Å. One RFdiffusion output is therefore one stochastic proposal, not a unique solution.

4 ProteinMPNN: sequence design on a fixed backbone

ProteinMPNN addresses the next question: given fixed backbone coordinates, which amino-acid identities are geometrically compatible with them [2]? It constructs a residue graph from backbone geometry, passes messages using invariant geometric features, and samples a sequence autoregressively. Schematically,

$$p(\mathbf{s} \mid X) = \prod_{k=1}^{L} p\big(s_{\pi_k} \mid s_{\pi_1}, \ldots, s_{\pi_{k-1}}, X\big)\,,$$

where X is the fixed backbone, $\mathbf{s}$ is the sequence, and π is a decoding order. Target residues can be fixed while only the generated binder chain is designed.

The model returns amino-acid probabilities and a sequence-model score. A low negative log-probability means that ProteinMPNN considers the sequence compatible with the supplied backbone under its learned distribution. It does not mean low binding free energy, high affinity, or high experimental success.

For the fixed 90-residue RFdiffusion seed-42 backbone, the audited run generated 100 unique ProteinMPNN sequences at temperature 0.1. Median pairwise identity was 66.7%, median position entropy was 0.648 nats, and 17 of 90 positions were invariant in this finite sample. These data provide a useful sequence ensemble for studying which backbone positions are tightly constrained, but they do not yet rank binding.

5 Where a physics-aware score could enter

The cleanest initial use of a new interface score is beside ProteinMPNN, after RFdiffusion has fixed the backbone:

$$\begin{aligned}\text{RFdiffusion backbones} &\longrightarrow \text{ProteinMPNN sequence ensemble}\\ &\longrightarrow \text{all-atom reconstruction and interface scoring}\\ &\longrightarrow \text{diverse shortlist.}\end{aligned}$$

This preserves the interpretation of each score. ProteinMPNN likelihood asks whether a sequence is compatible with a backbone; the physical score asks whether the resulting interface has favorable chemistry and geometry. An initial score could include clash penalties, buried nonpolar surface, unsatisfied polar groups, hydrogen-bond geometry, electrostatic complementarity, and penalties for exposed hydrophobes.

Three levels of integration should be distinguished:

3. **Reranking:** generate sequences normally, reconstruct side chains, and rank with the physical score. This is simplest and most auditable.

2. **Biased sampling:** combine ProteinMPNN log probability with a differentiable or locally updated interface score during sequence sampling. This can improve efficiency but requires calibration of two unlike scores.

3. **Joint retraining:** train or fine-tune a sequence model with the physical objective. This is the most ambitious option and needs reliable labels, negative examples, and controls against score exploitation.

For a starter study, reranking is preferable because it lets us test whether the physical score adds information beyond ProteinMPNN likelihood without changing either pretrained model.

6 Planned Week 35 deliverables

1. Trace one TYK2–4GIH forward pass through the released Boltz-2, Nesso-1, and FlashBind implementations and identify the exact information consumed by each affinity head.

2. Add one architecture-matched perturbation for each affinity model, as defined in the Week 34 report.

3. Present the RFdiffusion target, contig, hotspot tensor, noisy binder frames, and final backbone as an explicit input–output diagram.

4. Present the ProteinMPNN residue graph, design mask, logits, sampled sequences, and position-wise entropy for the fixed insulin-receptor binder backbone.

5. Define a small, prespecified physics-aware reranking score and test whether it selects information different from ProteinMPNN likelihood. No claim of affinity improvement will be made without an independent structural or experimental test.

7 Evidence boundary

The insulin-receptor calculations currently establish execution, conditioning behavior, stochastic structural diversity, and sequence diversity. They do not establish that any generated sequence folds, binds the receptor, modulates signaling, or has measurable affinity. The first useful scientific advance is therefore not simply generating more candidates; it is defining an independent and falsifiable selection procedure.

References

[1] Joseph L. Watson et al. De novo design of protein structure and function with RFdiffusion. *Nature*, 620:1089–1100, 2023. doi: 10.1038/s41586-023-06415-8. URL `https://doi.org/10.1038/s41586-023-06415-8`.

[2] Justas Dauparas et al. Robust deep learning–based protein sequence design using ProteinMPNN. *Science*, 378(6615):49–56, 2022. doi: 10.1126/science.add2187. URL `https://doi.org/10.1126/science.add2187`.